Directional Consistency in Fine-Grained NLI: Problem Formulation, Exploratory Analysis and Baselines for DiCo-NLI (Track 1)

CS-878 Deep Learning, Assignment 1

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1 Introduction and Task Definition

Natural Language Inference (NLI) is a standard test of whether models can identify semantic relations between texts. A model can score well on NLI benchmarks by exploiting superficial cues in the data (Gururangan et al., 2018). Standard evaluation also scores each ordered pair on its own, so it cannot show whether a model's judgements stay coherent when the same pair is presented in the opposite order.

DiCo-NLI (SemEval-2027 Task 2) addresses the second problem (DiCo-NLI Organizers, 2026). Systems classify the relation between two short phrases, and every reversible pair is evaluated in both orders, so classification quality and directional consistency are measured together (Apaolaza Larraya et al., 2026).

In this assignment we study Track 1 (English). We (i) analyse the data to identify task-relevant cues and potential shortcuts, (ii) build classical baselines, including deliberately restricted probes that use only shallow structural or lexical information, and (iii) fine-tune two pretrained encoders, BERT and RoBERTa. All systems are scored with the official metrics: Weighted F1, SoftCons and HardCons.

Research question. How far can shallow structural cues and standard fine-tuned encoders progress on DiCo-NLI, and to what extent does directional consistency reflect correct relational classification?

Hypotheses. **H1:** Surface cues such as phrase length yield non-trivial classification performance and high SoftCons. **H2:** Fine-tuned cross-encoders outperform the classical baselines on Weighted F1 and HardCons. **H3:** The remaining errors of the fine-tuned encoders concentrate at the boundary with NEG rather than on inference direction.

1.1 Task definition: black-box view

Each instance is an ordered pair of short phrases $x = (t_1, t_2)$, and the system predicts one of four labels: `EQUIVALENCE`, `FORWARD_ENTAILMENT`, `BACKWARD_ENTAILMENT` and `NEGATIVE_OTHER` (DiCo-NLI Organizers, 2026), abbreviated in the rest of this report as $\mathcal{Y} = \{\text{EQ}, \text{FWD}, \text{BWD}, \text{NEG}\}$. The labels can be read as answers to two questions: does t_1 entail t_2, and does t_2 entail t_1? EQ holds when $t_1 \Rightarrow t_2$ and $t_2 \Rightarrow t_1$; FWD when $t_1 \Rightarrow t_2$ but $t_2 \not\Rightarrow t_1$; BWD when $t_2 \Rightarrow t_1$ but $t_1 \not\Rightarrow t_2$. NEG groups all remaining relations of the source benchmark (opposition, similarity, relatedness and unrelatedness) into a single label (DiCo-NLI Organizers, 2026).

The direction of the input is therefore part of the prediction. For example, the training pair "Japanese PM Shinzo Abe" → "Abe" is labelled FWD; reversing the two phrases yields BWD. An equivalent pair such as "a monochrome photograph" and "a black and white photo" remains EQ after reversal, whereas "in southern Iraq" and "in northern Iraq" belong to NEG.

Formally, for the reversible labels we define the reversal operator

$$\mathrm{rev}(\textsc{Fwd}) = \textsc{Bwd}, \qquad \mathrm{rev}(\textsc{Bwd}) = \textsc{Fwd}, \qquad \mathrm{rev}(\textsc{Eq}) = \textsc{Eq}. \tag{1}$$

A directionally coherent classifier f should satisfy, for every reversible pair,

$$f(t_2, t_1) = \mathrm{rev}\big(f(t_1, t_2)\big). \tag{2}$$

NEG instances occur in one orientation only and are excluded from the pair-level consistency metrics; however, predicting NEG for either member of a reversible pair breaks Equation (2) and is counted as inconsistent.

1.2 Task definition: white-box view

Although the output is a four-class label, several properties make the task non-trivial. First, FWD and BWD differ only in direction. Symmetric signals such as lexical overlap or cosine similarity are unchanged when t_1 and t_2 are swapped and cannot, on their own, separate these labels. Second, high similarity does not imply equivalence: adding a modifier preserves much lexical and semantic similarity while changing the relation (e.g. "A sheep" vs. "A large sheep"). Third, NEG merges several different relations and is therefore internally heterogeneous. Fourth, phrases are very short, so individual words and modifiers carry most of the signal and little context is available. Finally, the data may contain exploitable structural cues: a more specific phrase often contains extra modifiers and is therefore longer than the phrase it entails, so a classifier could exploit length or containment patterns without modelling the relation. We test these potential shortcuts explicitly in the exploratory analysis (Section 4) and with restricted baselines (Section 6).

2 Related Work

Shortcuts in NLI data. Gururangan et al. (2018) showed that NLI datasets contain annotation artifacts: classifiers that see only the hypothesis reach about 67% accuracy on SNLI and 53% on MultiNLI, far above the 33% chance level. For this reason we include restricted diagnostic baselines that measure how far simple cues such as phrase length and lexical overlap go on their own.

Phrase-level inference data. DiCo-NLI is derived from PhrasIS (Lopez-Gazpio et al., 2024), a benchmark of over 10,000 naturally occurring phrase pairs from image captions and news headlines, annotated by experts for inference relations and semantic similarity. PhrasIS builds on the interpretable semantic textual similarity line of work, in which sentence pairs are decomposed into aligned chunks (Lopez-Gazpio et al., 2017), and evaluates semantic representations at a finer granularity than sentence-level NLI. At this granularity, small lexical or structural differences between short phrases can change the direction of inference. DiCo-NLI retains the reversible relations EQ, FWD and BWD, and maps all other PhrasIS relations to NEG (DiCo-NLI Organizers, 2026). NEG is therefore a comparatively heterogeneous category, which may make it harder to separate from the positive relations.

Directionality and the Reversal Curse. Berglund et al. (2024) showed that autoregressive language models trained on a fact stated in one direction ("A is B") often fail to answer the reversed query ("B is A"), so directional generalisation cannot be taken for granted. Wu et al. (2024) report that this behaviour depends on architecture: in their setting the bidirectional encoder BERT was immune to the reversal curse, although both BERT- and GPT-style models struggled with more complex deductive reasoning over three sets. The Reversal Curse concerns reverse retrieval of learned factual associations and is therefore related to, but distinct from, DiCo-NLI (DiCo-NLI Organizers, 2026), which asks whether the predicted relation for an ordered phrase pair transforms correctly when the order is reversed.

Coherence-based evaluation. The most direct predecessor of DiCo-NLI is the pilot study of Apaolaza Larraya et al. (2026), which evaluates logical coherence under reversed inference direction on PhrasIS phrase pairs. They introduce soft and hard causal coherence (SoftCoh and HardCoh): soft coherence measures whether predictions for the two orientations are mutually compatible, while hard coherence additionally requires both predictions to be correct. DiCo-NLI adopts these measures as SoftCons and HardCons. Across 17 encoder, decoder and encoder–decoder Transformers, encoders (in particular DeBERTa-v3) were strongest, and coherence fell by roughly 0.10–0.20 for encoders from seen to unseen phrase pairs. Errors were linked to surface similarity and lexical overlap and to prepositions. The authors also note that coherence does not imply correctness, because a model can be consistently wrong in both directions, and that gold labels can be ambiguous.

Positioning. Existing work evaluates directional coherence on PhrasIS with extensively tuned pretrained models. We examine how far simple structural, lexical and standard fine-tuned encoder baselines get on the official DiCo-NLI split, and we report classification performance alongside directional consistency so that reliance on shallow cues becomes visible.

3 Dataset

3.1 Data and splits

We use the official English data of Track 1, derived from PhrasIS (DiCo-NLI Organizers, 2026). The training set contains 3,042 rows corresponding to 1,760 underlying pairs (`pair_id`s), and the development set contains 660 rows corresponding to 383 pairs. The reference files additionally provide `reverse_pair_id`, which links the two orientations of each reversible pair and is used to compute the consistency metrics. We use the official splits unchanged. The trial data is not used because it overlaps with the training and development data.

3.2 Integrity checks

No missing or empty texts were found, and no `pair_id` occurs in both training and development. Otherwise one orientation of a pair could sit in training and its reversal in development, which would leak the answer. Across training and development combined, 116 rows are duplicated, and only one pair carries conflicting labels. Seven development rows have an exact ordered (t_1, t_2) match in the training data despite the absence of `pair_id` overlap; we keep the official split but treat this small overlap as a limitation.

3.3 Label distribution

The label distribution is nearly identical in both splits: FWD and BWD each account for about 29% of rows, EQ for about 26% and NEG for about 16%. FWD and BWD are balanced by construction, because each reversible entailment pair appears in both orientations, whereas NEG

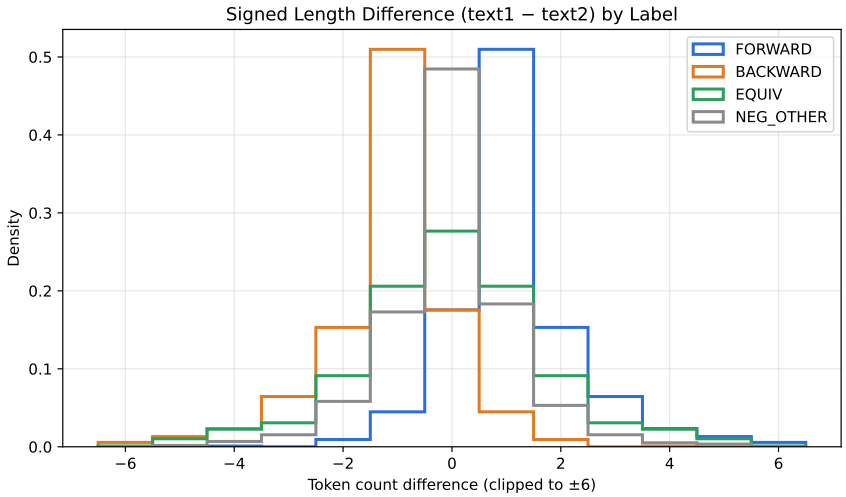


Figure 1: Distribution of the signed token-length difference $L(t_1) - L(t_2)$ per label (train and development). FWD and BWD are mirror images; EQ and NEG are centred on zero.

rows have no reversed counterpart. The imbalance is moderate, but because NEG is the smallest class we report Macro F1 alongside the official Weighted F1, so that performance on the less frequent class remains visible.

4 Exploratory Data Analysis

We analyse a small set of task-relevant feature groups: phrase length, lexical overlap and containment, part-of-speech (POS) patterns, semantic similarity, and vocabulary coverage. For each group we separate the observation from its interpretation and its modelling implication. All selected plots are provided in a separate EDA PDF; the two most informative are reproduced in Figures 1 and 2.

4.1 Phrase length and direction

Observation. For FWD, t_1 contains on average 3.58 tokens versus 2.44 for t_2, and BWD mirrors this pattern. About 77% of FWD rows place the longer phrase first, with the reverse for BWD; EQ and NEG have a mean signed length difference of zero (Figure 1). About 19% of rows contain at least one single-word phrase. **Interpretation.** A more specific phrase often carries additional lexical material and is longer than the more general phrase it entails. Length carries information about direction, although it is only a surface cue. **Modelling implication.** Signed length difference should be tested explicitly as a potential shortcut: Section 5 introduces directional length features and Section 6 evaluates a length-only baseline.

4.2 Lexical overlap and containment

Observation. In about 45% of entailment rows the word set of one phrase is fully contained in the other (Figure 2). Mean Jaccard similarity is about 0.41 for FWD and BWD, 0.27 for NEG and only 0.24 for EQ. **Interpretation.** Entailment often arises when one phrase adds information to the other, so directional containment is informative. Equivalent phrases, by contrast, frequently express the same meaning with different words. Moreover, Jaccard similarity is symmetric, $J(t_1, t_2) = J(t_2, t_1)$, so it cannot by itself distinguish FWD from BWD. **Modelling implication.** We separate directional containment features from symmetric overlap features and evaluate a symmetric-overlap-only probe.

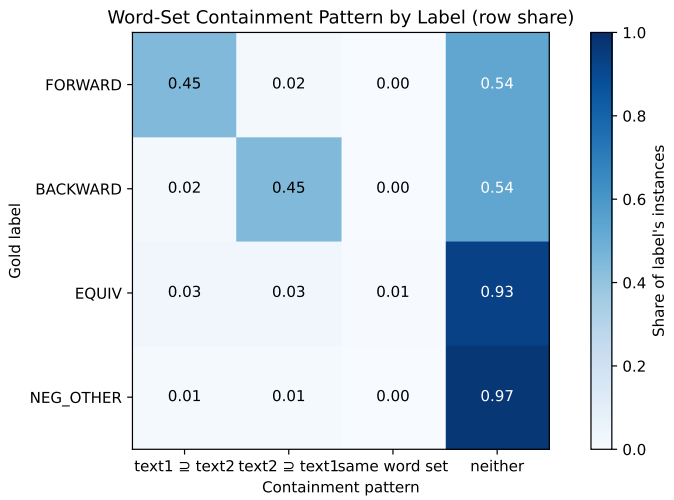


Figure 2: Word-set containment pattern per label (row-normalised). One-sided containment is characteristic of FWD and BWD; EQ and NEG pairs rarely contain one another.

4.3 Part-of-speech patterns

We examine the POS of words that occur in only one side of a pair. **Observation.** Compared with the other labels, words found only in t_1 of FWD pairs are especially often adjectives (0.39 per pair vs. 0.16 for EQ) and proper nouns (0.24 vs. 0.11), with the mirror pattern in t_2 for BWD. EQ and NEG pairs also contain many non-shared nouns (about 0.57 per side), but on both sides, which is characteristic of lexical substitution rather than one-sided addition. Across all tokens, NEG phrases also contain the highest proportion of prepositions (17.4%, compared with 12.9% for the entailment classes), consistent with contrastive pairs such as "to hospital" / "from hospital" (a NEG training pair). **Interpretation.** Adjectives and named-entity detail make a phrase more specific while preserving entailment of the more general phrase, so their directional distribution carries information about direction. **Modelling implication.** We add directional POS counts of non-shared words to the engineered features.

4.4 Semantic similarity

Each phrase is encoded independently with a frozen MiniLM sentence encoder (`all-MiniLM-L6-v2`) and the cosine similarity of the two embeddings is computed. **Observation.** Mean cosine similarity is about 0.74 for FWD and BWD, 0.72 for EQ and 0.62 for NEG. **Interpretation.** Similarity separates related from unrelated pairs, but it does not identify the direction or the exact type of relation. Cosine similarity is symmetric, $\cos(\mathbf{e}_1, \mathbf{e}_2) = \cos(\mathbf{e}_2, \mathbf{e}_1)$, and therefore direction-blind; the near-identical values for EQ and the entailment labels also show that similarity is not the same as inference. **Modelling implication.** A model needs to read the ordered pair jointly, so our pretrained baselines are pair encoders.

4.5 Vocabulary sparsity

Observation. About 47% of training vocabulary types occur only once, and 15.4% of development tokens are out of vocabulary with respect to the training data. **Interpretation.** Representations built only from training-set word identities have little or no evidence for rare and unseen words. **Modelling implication.** We compare a TF–IDF lexical baseline with pretrained encoders, whose subword tokenisation and pretraining provide information about such words.

Summary. The data contains both semantic structure and potentially exploitable surface cues. Directional length, containment, adjectives and proper nouns correlate with entailment direction, whereas Jaccard overlap and cosine similarity are symmetric and cannot separate FWD from BWD on their own. These findings determine the features and diagnostic baselines in Sections 5 and 6.

5 Feature Engineering

Guided by the EDA, we construct 26 numerical features for each ordered pair. Besides providing a classical baseline representation, they are designed to separate directional signals from symmetric ones.

Length (4). Token counts of t_1 and t_2, the signed token-length difference $L(t_1)-L(t_2)$, and the signed character-length difference. The signed differences change sign when the pair is reversed.

Symmetric overlap (2). Jaccard similarity of the two word sets and an indicator for identical texts. Both are unchanged under reversal.

Directional coverage and containment (6). The proportion of t_1's words found in t_2 and vice versa, binary full-containment indicators in both directions, and the numbers of words occurring only in t_1 and only in t_2. Unlike Jaccard similarity, these features indicate which phrase contains the additional material.

Task-specific cues (2). Negation mismatch is 1 when exactly one phrase contains a negation term (*not, no, never, n't, without, none, nobody, nothing*). Number mismatch is 1 when the sets of digit tokens differ. Because it considers digit tokens only, it does not normalise written and numeric forms such as "Two" and "2"; this is a limitation of the deliberately simple feature.

POS of non-shared words (12). Using spaCy (`en_core_web_sm`), we count NOUN, PROPN, ADJ, VERB, ADP and NUM tokens among the words that occur in only one phrase, separately for t_1 and t_2.

Directionality and preprocessing. Most features are directional: under reversal, signed quantities change sign and phrase-specific features exchange roles. Only Jaccard similarity and the identical-text indicator are symmetric; Section 6 exploits this distinction through separate diagnostic probes. The 26 features are standardised with z-score parameters fitted on the training data, which also makes the logistic-regression coefficients comparable. The MiniLM similarity of Section 4 is used only for exploration and is not a classifier input.

6 Baseline Models

We evaluate a progression from deliberately weak diagnostic models to fully fine-tuned pretrained encoders, in order to measure how much performance simple dataset cues provide before richer representations are introduced.

6.1 Baseline A: classical models

All logistic-regression (LR) classifiers use scikit-learn with at most 5,000 iterations, the default regularisation $C = 1$, and a fixed random state.

- **Majority.** Predicts the most frequent training label. FWD and BWD are tied at 881 training rows; the implementation resolves the tie to FWD.
- **LR, length only.** Uses only the four length features, to quantify how much classification and consistency the length cue alone provides.
- **LR, symmetric overlap only.** Uses only Jaccard similarity and the identical-text indicator. Since both satisfy $g(t_1, t_2) = g(t_2, t_1)$, this model has no representation of direction and serves as a direction-blind probe.
- **LR, engineered features.** Uses all 26 standardised features.
- **TF–IDF + LR.** Unigram and bigram TF–IDF with sublinear term frequency, with the vocabulary fitted on training phrases only. The two phrases are vectorised separately and concatenated, $\mathbf{x} = [\text{tfidf}(t_1) \,\|\, \text{tfidf}(t_2)]$, so that phrase order is retained.
- **TF–IDF + features + LR.** Concatenates the TF–IDF representation with the 26 standardised features.

6.2 Baseline B: fine-tuned pretrained encoders

We fine-tune `bert-base-uncased` and `roberta-base`, each with a new four-way classification head; all encoder weights are updated (no frozen features). Both models are cross-encoders that read the two phrases jointly and in order: BERT receives [CLS] t_1 [SEP] t_2 [SEP] with segment IDs (`token_type_ids`) marking the two phrases, while RoBERTa uses $\langle s\rangle\, t_1\, \langle /s\rangle\langle /s\rangle\, t_2\, \langle /s\rangle$ without segment IDs. Unlike symmetric similarity measures, these encoders can learn order-dependent representations.

The maximum sequence length is 64 tokens and the batch size 16. We use AdamW with learning rate 3×10^{-5}, weight decay 0.01, gradient clipping at 1.0 and a linear schedule with 10% warm-up, and train for six epochs. The final epoch is used, a rule fixed in advance; there is no early stopping and no selection of the best development epoch. Each model is trained with three seeds (13, 42, 87) on an Apple M4 GPU via PyTorch's MPS backend, and we report the mean and standard deviation.

Implementation note. In a preliminary run the BERT batching pipeline omitted `token_type_ids`, and the training schedule (four epochs, learning rate 2×10^{-5}) left the training loss far from converged. Both issues were corrected once, after which the configuration above was frozen. The decision was based on the implementation error and the training loss, not on development-set scores.

7 Evaluation Setup

All systems are evaluated on the official development set (660 rows) with the organisers' scorer. Of these rows, 554 form $M = 277$ reversible pairs (190 FWD/BWD pairs and 87 EQ pairs), and 106 are NEG rows without a reversed counterpart. We evaluate row-level classification separately from pair-level directional consistency.

7.1 Classification metrics

For each class c, precision, recall and F1 are

$$P_c = \frac{TP_c}{TP_c + FP_c}, \qquad R_c = \frac{TP_c}{TP_c + FN_c}, \qquad F1_c = \frac{2P_cR_c}{P_c + R_c}. \tag{3}$$

The official classification metric is Weighted F1,

$$\text{W-F1} = \sum_{c \in \mathcal{Y}} \frac{n_c}{N} F1_c, \tag{4}$$

where n_c is the number of development rows with gold label c and $N = 660$. Each class contributes in proportion to its support: NEG receives a weight of about 0.16 and the other labels about 0.26–0.29 each. We also report Macro F1, $\frac{1}{|\mathcal{Y}|}\sum_c F1_c$, which gives every class a weight of 0.25. Because NEG is both the smallest class and empirically a difficult one, Weighted F1 gives it less influence than Macro F1 does, and the gap between the two reveals how much a model struggles on this class. Accuracy is reported as a complementary statistic.

7.2 SoftCons

Let $\hat{y}_{i,1}$ and $\hat{y}_{i,2}$ be the predictions for the two orientations of reversible pair i. Then

$$\text{SoftCons} = \frac{1}{M}\sum_{i=1}^{M} \mathbb{1}[\hat{y}_{i,1} \in \{\text{EQ}, \text{FWD}, \text{BWD}\} \ \wedge \ \hat{y}_{i,2} = \text{rev}(\hat{y}_{i,1})]\,. \tag{5}$$

SoftCons does not use the gold labels: it measures whether a model's two predictions are compatible, not whether either is correct. A NEG prediction on either orientation automatically fails the condition.

For example, consider the EQ pair "same-sex marriage" and "gay marriage". The first phrase is longer, and the length-only classifier predicts FWD in one orientation and BWD after reversal. These predictions satisfy the reversal mapping and therefore count as consistent under SoftCons, yet both are incorrect because the gold relation is EQ in both orientations. A classifier can thus exploit a directional surface cue to achieve high consistency without correspondingly accurate classification.

7.3 HardCons

HardCons additionally requires both orientations to be correct:

$$\text{HardCons} = \frac{1}{M}\sum_{i=1}^{M} \mathbb{1}[\hat{y}_{i,1} = y_{i,1} \ \wedge \ \hat{y}_{i,2} = y_{i,2}]\,. \tag{6}$$

Because the gold labels of reversible pairs themselves satisfy the reversal mapping, any pair counted by HardCons is also counted by SoftCons, so HardCons $\leq$ SoftCons. SoftCons asks whether a model behaves coherently under reversal, HardCons whether it is coherently correct, and Weighted and Macro F1 how well it classifies individual rows. We report all of them because a model can be highly consistent and still often wrong.

8 Results

Table 1 reports all systems on the development set. Classical models are deterministic single runs; the fine-tuned encoders are averaged over three seeds.

Classical baselines. The majority baseline predicts FWD for every row, so every flipped twin is wrong and both consistency scores are zero. Using only four length features, LR reaches 0.464 W-F1 and an almost perfect SoftCons of 0.989, while its HardCons is only 0.628. The symmetric-overlap probe, which sees identical inputs for both orientations, reaches only 0.324 W-F1 and 0.368 SoftCons: it can be consistent only by predicting EQ in both directions. Adding directional features raises W-F1 to 0.593, TF–IDF alone reaches 0.484, and their combination is the strongest classical model (0.620 W-F1, 0.661 HardCons). Lexical identity alone is therefore weaker than features that compare the two phrases.

Model	W-F1	SoftCons	HardCons	Macro-F1	Acc.
Baseline A: classical					
Majority class	0.129	0.000	0.000	0.112	0.288
LR (length only)	0.464	**0.989**	0.628	0.408	0.532
LR (symmetric overlap only)	0.324	0.368	0.188	0.291	0.370
LR (engineered features)	0.593	0.877	0.653	0.558	0.602
TF–IDF + LR	0.484	0.877	0.542	0.452	0.500
TF–IDF + features + LR	0.620	0.856	0.661	0.588	0.626
Baseline B: fine-tuned encoders (mean ± std over 3 seeds)					
BERT-base	0.705 ± 0.023	0.774 ± 0.016	0.693 ± 0.021	0.680 ± 0.024	0.707 ± 0.020
RoBERTa-base	**0.765** ± 0.002	0.804 ± 0.011	**0.739** ± 0.006	**0.742** ± 0.002	**0.762** ± 0.004

Table 1: Development-set results (Track 1, English) computed with the official scorer. W-F1, SoftCons and HardCons are the official metrics.

Model	FWD	BWD	EQ	NEG
TF–IDF + features + LR	0.744	0.739	0.515	0.355
BERT-base (3 seeds)	0.813	0.816	0.587	0.503
RoBERTa-base (3 seeds)	0.839	0.839	0.725	0.565

Table 2: Per-class F1 on the development set (encoders: mean over three seeds).

Coefficients. The standardised coefficients of the engineered-feature model agree with the EDA. FWD is driven by non-shared proper nouns, nouns and adjectives in t_1, together with a positive character-length difference, and is penalised by non-shared material in t_2; BWD shows the mirror image. EQ has positive weights when both phrases contain non-shared words, the substitution pattern of paraphrases, and negative weights for non-shared numerals. NEG is most strongly penalised by full containment in either direction. Because the length and non-shared-word features are correlated, individual coefficients should be interpreted jointly rather than in isolation.

Fine-tuned encoders. Both encoders outperform the classical baselines. RoBERTa reaches 0.765 ± 0.002 W-F1 and 0.739 ± 0.006 HardCons, compared with 0.705 ± 0.023 and 0.693 ± 0.021 for BERT. The gap of 0.06 W-F1 is well above the seed standard deviations, and RoBERTa is also much more stable across seeds. Per-class F1 (Table 2) shows that the largest improvement occurs for EQ, which rises from 0.515 for the best classical model to 0.725 for RoBERTa. This matches the EDA: equivalent phrases share few words (Jaccard 0.24), so recognising them requires lexical knowledge that pretraining provides. NEG remains the weakest class for every model.

Consistency versus correctness. The encoders are *less* consistent than the classical models (SoftCons 0.774–0.804 vs. 0.856–0.877) but *more* often correct in both directions (HardCons 0.693–0.739 vs. at most 0.661). Together with the length-only result, this shows that SoftCons and correctness can move in opposite directions.

Training dynamics. Training loss falls to about 0.06–0.12 by the sixth epoch, while development W-F1 levels off after three to four epochs (Figure 3). The best development epoch was typically the fifth, within 0.01–0.02 W-F1 of the final epoch, so using the pre-specified final epoch costs little; the widening gap between training loss and development performance indicates mild overfitting. HardCons fluctuates more between epochs than W-F1 (e.g. RoBERTa, seed 42: 0.769 at epoch 4 and 0.736 at epoch 6), indicating that pair-level metrics are noisier.

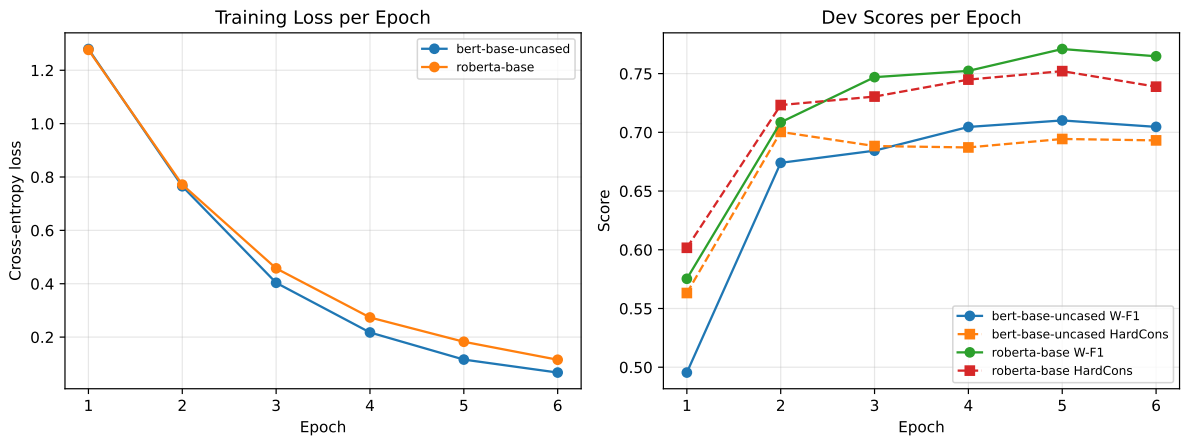


Figure 3: Training loss (left) and development W-F1 and HardCons (right) per epoch, averaged over three seeds. Curves are diagnostic only and were not used for model selection.

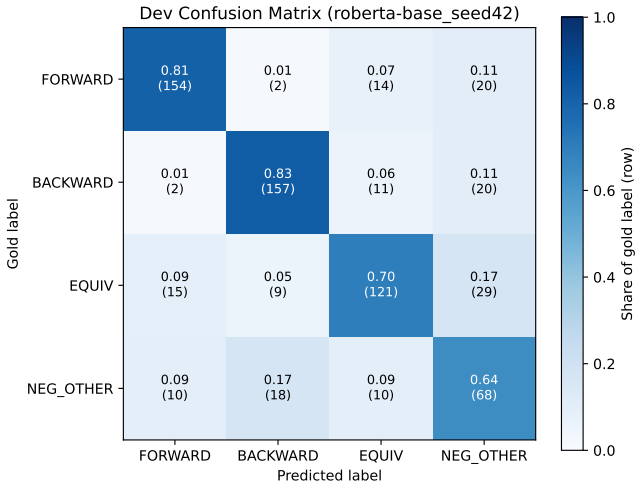


Figure 4: Row-normalised development confusion matrix of RoBERTa-base (seed 42); counts in parentheses.

Context. On the public development-phase leaderboard of the task, the strongest Track 1 entries report W-F1 above 0.95 with SoftCons of 1.0, and the third entry reports 0.808 / 0.906 / 0.827. Our RoBERTa baseline (0.765 / 0.804 / 0.739) is below these but well above the fourth entry (0.510). This comparison is indicative only: development labels are public, so leaderboard systems may have been trained or tuned on them.

9 Error Analysis

We analyse seed 42 of each encoder and the strongest classical model.

9.1 Confusion patterns

Direction is largely handled by the encoders: both BERT and RoBERTa confuse FWD with BWD only twice in each direction, compared with 7 and 9 cases for the classical model. RoBERTa's errors concentrate at the boundary with NEG (Figure 4): 20 FWD and 20 BWD rows are predicted as NEG, 29 EQ rows are predicted as NEG, and 18 NEG rows are predicted as BWD.

Outcome	TF–IDF + feats	BERT	RoBERTa
Both directions correct	183	186	**204**
Consistent but wrong	**54**	28	16
Direction-blind (same entailment label both ways)	4	4	0
NEG predicted in one direction only	35	47	42
Other inconsistency	1	12	15

Table 3: Outcomes for the 277 reversible development pairs (seed 42 for the encoders).

Heuristic error category	TF–IDF + feats	RoBERTa
Related pair rejected as NEG	56	**69**
Followed length cue (longer side entails)	68	38
EQ vs. entailment confusion	**80**	34
Contrastive/unrelated pair predicted as related	43	19

Table 4: Heuristic error categories (misclassified development rows).

9.2 Pair-level outcomes

Table 3 classifies the 277 reversible pairs by the joint outcome of their two predictions.

The classical model owes much of its consistency to being consistently wrong (54 pairs), in line with the length-cue effect. The encoders produce far fewer such pairs; their main source of inconsistency is a NEG prediction in only one orientation (42 pairs for RoBERTa), which automatically violates SoftCons and explains why their SoftCons is lower despite higher HardCons.

9.3 Error categories

We tag every misclassified row with a heuristic category (Table 4). For the classical model the largest category is confusion between EQ and an entailment label (80 rows), followed by predictions that follow the length cue (68). For RoBERTa, length-cue errors fall to 38, a reduction of 44%, and EQ–entailment confusions to 34; the largest category becomes related pairs rejected as NEG (69 rows).

9.4 Qualitative examples

The EQ pair "a monochrome photograph" / "a black and white photo" shows different behaviour across models. The length-only classifier predicts EQ in both orientations, plausibly because the two phrases have the same number of characters (the signed character difference is zero). TF–IDF plus features predicts BWD and FWD, a consistent-but-wrong pair; BERT predicts EQ in one direction and NEG in the other; RoBERTa correctly predicts EQ in both. The pair "same-sex marriage" / "gay marriage" (gold EQ) is FWD/BWD for the length-only model, the best classical model and RoBERTa alike, and FWD/FWD for BERT.

Some RoBERTa errors involve distinguishing equivalence from one-way entailment. "A black car" → "A black vehicle" is labelled FWD but predicted EQ: the phrases are closely related, but a car is a vehicle whereas a vehicle need not be a car. Conversely, "on green grass" / "on a lawn" is labelled EQ but predicted FWD.

In the most frequent RoBERTa category, related pairs are rejected as NEG, for example "martial arts clubs" / "Martial Arts Schools" and "to pay off" / "of bribing", both labelled EQ. These near-paraphrases with different surface forms illustrate that the main remaining difficulty lies at the boundary between the reversible relations and NEG.

Prepositional variation also remains challenging, echoing the findings of Apaolaza Larraya et al. (2026). RoBERTa predicts NEG for "from Japan tunnel" / "in collapsed Japan tunnel"

(gold BWD), and BERT predicts BWD for "out of the water" / "through the water" (gold NEG).

Finally, some disagreements expose the ambiguity of fine-grained labels. "Australian senator" / "Australian lawmaker" is labelled NEG, although a FWD reading is arguably defensible because a senator is a type of lawmaker. Cases like this limit what any single disagreement can tell us about a model.

9.5 Segment information in BERT

The preliminary BERT run without `token_type_ids` assigned the same entailment label to both orientations in 119 of the 190 FWD/BWD pairs, and its HardCons was 0.28. With segment IDs restored, the number of such direction-blind pairs falls to zero. Together with the revised training schedule, HardCons rose to 0.693 ± 0.021. The disappearance of direction-blind predictions is directly attributable to the segment information: BERT relies on segment embeddings to know which phrase comes first, whereas RoBERTa encodes order through separator tokens and positions only.

9.6 Summary of failure modes

Shallow models gain consistency by following directional surface cues and are often consistently wrong. Fine-tuned encoders rarely confuse the two entailment directions. Most of their remaining errors involve the boundary with NEG, with fewer between equivalence and one-way entailment and around prepositions.

10 Discussion

Simple structural cues produce very high directional consistency without accurate classification. Fine-tuned encoders improve correctness considerably, and their remaining errors move from direction to the boundary with NEG.

H1 (supported). With four length features, LR achieves 0.464 W-F1 but 0.989 SoftCons, consistent with the EDA finding that about 77% of FWD rows place the longer phrase first. Signed length changes predictably under reversal, so the model produces compatible FWD/BWD pairs even when the relation is wrong, as for "same-sex marriage" / "gay marriage". The best classical model and RoBERTa make the same consistent-but-wrong prediction for this pair, and BERT predicts FWD in both directions. Richer representations make this failure much rarer (54 consistent-but-wrong pairs for the classical model, 16 for RoBERTa), yet hard equivalence cases remain. A high SoftCons on its own says little about classification quality.

H2 (supported). The best classical model reaches 0.620 W-F1 and 0.661 HardCons; RoBERTa reaches 0.765 ± 0.002 and 0.739 ± 0.006, and BERT 0.705 ± 0.023 and 0.693 ± 0.021. The largest gain is for EQ (F1 0.515 $\rightarrow$ 0.725), consistent with the low lexical overlap of equivalent pairs: recognising a paraphrase takes lexical knowledge that surface matching lacks and pretraining provides.

H3 (supported). FWD/BWD confusion becomes rare for the encoders (2 + 2 cases vs. 7 + 9 for the classical model). The dominant RoBERTa error is instead rejecting related pairs as NEG (69 rows), and 42 reversible pairs receive NEG in only one orientation.

Consistency versus correctness. Across our models, SoftCons and correctness move in opposite directions. The length-only model has 0.989 SoftCons with 0.628 HardCons; RoBERTa has 0.804 ± 0.011 SoftCons with 0.739 ± 0.006 HardCons. The pair-level analysis explains why:

the classical model produces 54 consistent-but-wrong pairs against 16 for RoBERTa, whereas RoBERTa's one-sided NEG predictions lower its SoftCons. SoftCons alone would rank the length-only model first. HardCons ties coherence to correctness, and row-level F1 covers all four labels, including NEG.

Limitations. (i) We evaluate on the development split rather than a blind test set; development labels are public, and seven development rows have exact text matches in training. (ii) No hyperparameter search was performed. The encoder configuration was fixed after correcting the preliminary implementation issue, which limits tuning on the development set but means the models may not reach their best achievable performance. (iii) MPS execution is not fully deterministic: repeated RoBERTa runs with the same seed and configuration differed by up to about 0.03 W-F1. We therefore report means over three seeds, although more seeds would give tighter estimates. (iv) Error categories are assigned by heuristic rules and are neither manually verified nor mutually exclusive. (v) A DeBERTa-v3 experiment was attempted but excluded because training produced NaN losses on the MPS backend. (vi) Some fine-grained gold labels are arguable, which limits conclusions drawn from individual disagreements.

11 Next Steps

1. **Joint prediction of reversed pairs.** Our classifiers predict each row independently, yet RoBERTa produces 42 reversible pairs with NEG in only one orientation. A joint decoder that scores both orientations together could enforce or encourage Equation (2) and remove this source of inconsistency.

2. **Reversal-aware training.** Training on paired orientations, or adding a consistency objective that encourages $f(t_2, t_1) = \mathrm{rev}(f(t_1, t_2))$, could improve consistency without sacrificing accuracy; reverse training (Golovneva et al., 2024) is a relevant starting point.

3. **Better modelling of NEG.** NEG remains the weakest class (RoBERTa F1 0.565) and the largest error category. Class-weighted objectives, or modelling its underlying PhrasIS relations explicitly, may help.

4. **Additional encoders and blind evaluation.** DeBERTa-v3 should be evaluated in a CUDA environment for a fairer comparison, and the final systems should be submitted to the official blind test phase to check whether development findings generalise.

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